

ECON 628

Problem Set #1 — Solutions

Fall 2026

Question 1: Identification — a structural labour supply model

(a) The payoffs to the two feasible hours choices are

$$\begin{aligned} u_{20} &= 20w_i - \frac{d_i}{1+a} 20^{1+a}, \\ u_{40} &= 40w_i - \frac{d_i}{1+a} 40^{1+a}. \end{aligned}$$

(b) The agent works 40 hours whenever $u_{40} > u_{20}$:

$$\begin{aligned} \mathbb{P}(h_i = 40 \mid w_i) &= \mathbb{P}\left(40w_i - \frac{d_i}{1+a} 40^{1+a} > 20w_i - \frac{d_i}{1+a} 20^{1+a} \mid w_i\right) \\ &= \mathbb{P}\left(20w_i > \frac{d_i}{1+a} (40^{1+a} - 20^{1+a}) \mid w_i\right) \\ &= \mathbb{P}\left(\log d_i < \log 20 + \log w_i + \log(1+a) - \log(40^{1+a} - 20^{1+a}) \mid w_i\right) \\ &= \Phi\left(\frac{\log 20 + \log w_i + \log(1+a) - \log(40^{1+a} - 20^{1+a}) - \mu}{\sigma}\right). \end{aligned}$$

It is convenient to name the wage-free part of the threshold

$$K(a) \equiv \log 20 + \log(1+a) - \log(40^{1+a} - 20^{1+a}),$$

so that

$$\mathbb{P}(h_i = 40 \mid w_i) = \Phi\left(\frac{K(a) - \mu}{\sigma} + \frac{1}{\sigma} \log w_i\right). \quad (1)$$

(c) With a binary choice set the individual likelihood contribution is

$$\begin{aligned} \mathcal{L}_i &= \mathbb{P}(h_i = 40 \mid w_i)^{\mathbf{1}\{h_i=40\}} \cdot \mathbb{P}(h_i = 20 \mid w_i)^{\mathbf{1}\{h_i=20\}} \\ &= \mathbb{P}(h_i = 40 \mid w_i)^{\mathbf{1}\{h_i=40\}} \cdot (1 - \mathbb{P}(h_i = 40 \mid w_i))^{\mathbf{1}\{h_i=20\}}. \end{aligned}$$

Letting $y_i = \mathbf{1}\{h_i = 40\}$, the sample log-likelihood is

$$\ell(\theta; Y, W) = \sum_{i=1}^n y_i \log \Phi\left(\frac{K(a) - \mu}{\sigma} + \frac{1}{\sigma} \log w_i\right) + (1 - y_i) \log \left[1 - \Phi\left(\frac{K(a) - \mu}{\sigma} + \frac{1}{\sigma} \log w_i\right)\right].$$

(d) This is a Maximum Likelihood estimator. Because $\log d_i \stackrel{iid}{\sim} N(\mu, \sigma)$, the multinomial choice model can be estimated with a probit.

(e) Write the probit regression as $y_i = \alpha + \beta \log w_i$ with

$$\alpha = \frac{K(a) - \mu}{\sigma}, \quad \beta = \frac{1}{\sigma}.$$

Only (α, β) are identified from the data, so:

- σ is identified as $1/\beta$, i.e. from the *slope* on $\log w_i$;
- given σ , the intercept identifies $K(a) - \mu$;
- a and μ are **not** separately identified.

(f) With choice set $\{10, 20, 40\}$ we need the payoff to the chosen option to beat both others. Start with $h = 40$. The two binding comparisons are

$$u_{40} > u_{20} \iff d_i < \frac{20w_i(1+a)}{40^{1+a} - 20^{1+a}}, \quad u_{40} > u_{10} \iff d_i < \frac{30w_i(1+a)}{40^{1+a} - 10^{1+a}}.$$

The first constraint is the tighter one for every $a \in (0, \infty)$, since

$$\log 20 - \log 30 - [\log(40^{1+a} - 20^{1+a}) - \log(40^{1+a} - 10^{1+a})] \leq 0,$$

with equality only at $a = 0$. Hence the $u_{40} > u_{10}$ comparison is slack and

$$\mathbb{P}(h_i = 40 \mid w_i) = \Phi\left(\frac{K(a) - \mu}{\sigma} + \frac{1}{\sigma} \log w_i\right), \quad (2)$$

which is *identical* to (1): adding the 10-hour option does not change the probability of working 40 hours.

For $h = 20$ we need $u_{20} > u_{10}$ and $u_{20} > u_{40}$, i.e. a two-sided restriction on d_i :

$$\frac{20w_i(1+a)}{40^{1+a} - 20^{1+a}} < d_i < \frac{10w_i(1+a)}{20^{1+a} - 10^{1+a}}.$$

Defining, in parallel with $K(a)$,

$$F(a) \equiv \log 10 + \log(1+a) - \log(20^{1+a} - 10^{1+a}),$$

this becomes

$$\mathbb{P}(h_i = 20 \mid w_i) = \Phi\left(\frac{F(a) - \mu}{\sigma} + \frac{\log w_i}{\sigma}\right) - \Phi\left(\frac{K(a) - \mu}{\sigma} + \frac{\log w_i}{\sigma}\right). \quad (3)$$

(g) Here everything is identified. After running the multinomial probit, we'll be able to first estimate σ and $K(a) - \mu$. We can do the same thing for the other equation and get $F(a) - \mu$ (intercept). The system of equations:

$$\begin{cases} \log 20 + \log(1+a) - \log(40^{1+a} - 20^{1+a}) - \mu = p, \\ \log 10 + \log(1+a) - \log(20^{1+a} - 10^{1+a}) - \mu = r, \end{cases}$$

Can be solved in (a, μ) .

Question 2: Iterated Projections

We want $\mathbb{E}^*[Y_i | X_i] = \mathbb{E}^*[\mathbb{E}^*[Y_i | X_i, Z_i] | X_i]$.

Define the projection error,

$$\varepsilon_i = Y_i - \mathbb{E}^*[Y_i | X_i, Z_i].$$

By construction, ε_i is orthogonal to everything it was projected on:

$$\mathbb{E}[\varepsilon_i] = 0, \quad \mathbb{E}[X_i \varepsilon_i] = 0, \quad \mathbb{E}[Z_i \varepsilon_i] = 0.$$

Now write $Y_i = \mathbb{E}^*[Y_i | X_i, Z_i] + \varepsilon_i$ and project both sides on X_i . Linear projection is a linear operator, so

$$\begin{aligned} \mathbb{E}^*[Y_i | X_i] &= \mathbb{E}^*[\mathbb{E}^*[Y_i | X_i, Z_i] + \varepsilon_i | X_i] \\ &= \mathbb{E}^*[\mathbb{E}^*[Y_i | X_i, Z_i] | X_i] + \mathbb{E}^*[\varepsilon_i | X_i]. \end{aligned}$$

The second term is 0: the linear projection of ε_i on X_i is the familiar OLS-like minimizer

$$\mathbb{E}^*[\varepsilon_i | X_i] = X_i' \mathbb{E}[X_i X_i']^{-1} \mathbb{E}[X_i \varepsilon_i] = 0,$$

because $\mathbb{E}[X_i \varepsilon_i] = 0$. Hence

$$\mathbb{E}^*[Y_i | X_i] = \mathbb{E}^*[\mathbb{E}^*[Y_i | X_i, Z_i] | X_i]. \quad \blacksquare$$

Question 3: FWL theorem

Let $\mathbb{E}^*[Y_i | X_i, Z_i] = X_i' \beta + Z_i' \gamma$, $\tilde{Z}_i = Z_i - \mathbb{E}^*[Z_i | X_i]$ and $\tilde{Y}_i = Y_i - \mathbb{E}^*[Y_i | X_i]$.

(a) From Q2, $\mathbb{E}^*[Y_i | X_i] = X_i' \beta + \mathbb{E}[Z_i' | X] \gamma$, so

$$Y_i - \mathbb{E}^*[Y_i | X_i] = Y_i - X_i' \beta - \mathbb{E}[Z_i' | X] \gamma \implies \tilde{Y}_i = \tilde{Z}_i' \gamma + \varepsilon_i.$$

Multiplying through by $\tilde{Z}_i$ and taking expectations, with $\mathbb{E}[\tilde{Z}_i \varepsilon_i] = 0$,

$$\mathbb{E}[\tilde{Z}_i \tilde{Y}_i] = \mathbb{E}[\tilde{Z}_i \tilde{Z}_i'] \gamma \implies \gamma = \mathbb{E}[\tilde{Z}_i \tilde{Z}_i']^{-1} \mathbb{E}[\tilde{Z}_i \tilde{Y}_i].$$

(b) By the law of iterated projections (Question 2) and linearity,

$$\mathbb{E}^*[Y_i | X_i] = \mathbb{E}^*[\mathbb{E}^*[Y_i | X_i, Z_i] | X_i] = \mathbb{E}^*[X_i' \beta + Z_i' \gamma | X_i] = X_i' \beta + \mathbb{E}^*[Z_i | X_i]' \gamma.$$

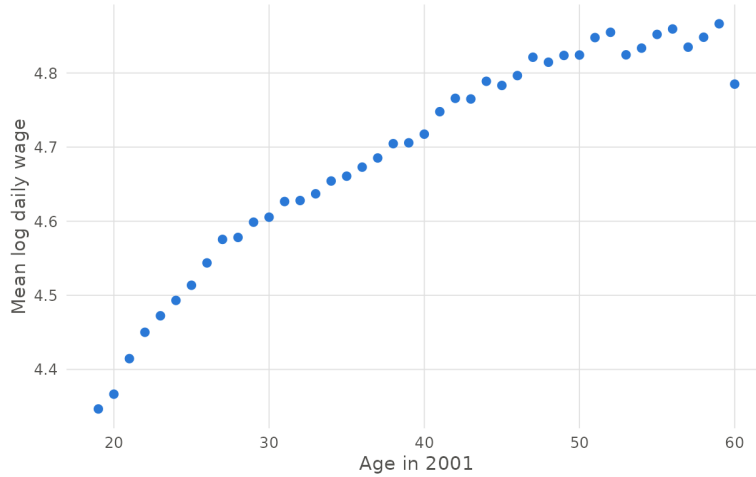
Now add and subtract $\mathbb{E}^*[Y_i | X_i]$:

$$\begin{aligned} Y_i - \mathbb{E}^*[Y_i | X_i, Z_i] &= Y_i - X_i' \beta - Z_i' \gamma \\ &= (Y_i - \mathbb{E}^*[Y_i | X_i]) + \underbrace{\mathbb{E}^*[Y_i | X_i] - X_i' \beta - Z_i' \gamma}_{= X_i' \beta + \mathbb{E}^*[Z_i | X_i]' \gamma} \\ &= (Y_i - \mathbb{E}^*[Y_i | X_i]) - (Z_i - \mathbb{E}^*[Z_i | X_i])' \gamma \\ &= \tilde{Y}_i - \tilde{Z}_i' \gamma. \quad \blacksquare \end{aligned}$$

Question 4: Saturated Models

(a)–(b) Load `veneto.csv` and build `femalei = 1{sexi = F}` and `agei = 2001 – birthi`.

(c) Mean log daily wage by age, ages 19–60.



Average log wage by age

(d) The raw profile in (c) is recovered exactly by a **saturated** regression on a full set of age

dummies:

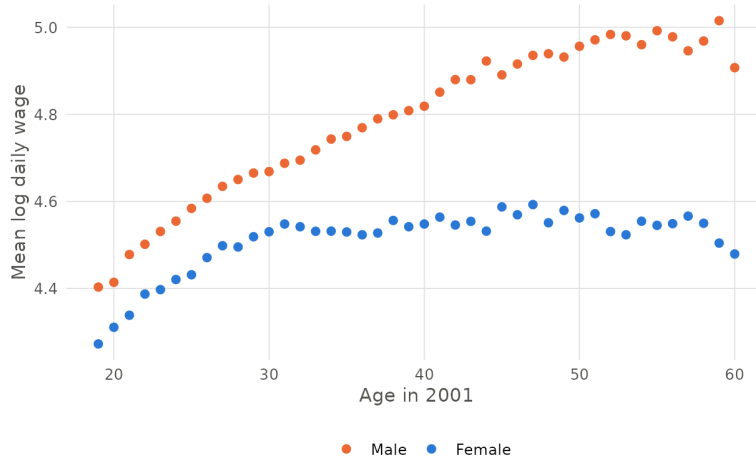
$$\log w_i = \sum_{k=19}^{60} \mathbf{1}\{\text{age}_i = k\} \pi_k + \varepsilon_i.$$

There is no constant (or, equivalently, drop one dummy and keep a constant). Because the regressor set is a partition of the support of age_i , the linear projection *coincides* with the conditional expectation function, and each coefficient is the corresponding cell mean:

$$\hat{\pi}_k = \frac{1}{N_k} \sum_{i: \text{age}_i = k} \log w_i = \widehat{\mathbb{E}}[\log w_i \mid \text{age}_i = k].$$

So the fitted values of this regression, plotted against age, reproduce the dots of panel (c) one for one: this is the sense in which a saturated regression is non-parametric.

(e)



Average log wage by age, female vs. male

(f) Both profiles come out of a *single* regression that saturates the $(\text{age} \times \text{female})$ cells:

$$\log w_i = \sum_k \sum_{g \in \{0,1\}} \mathbf{1}\{\text{age}_i = k\} \cdot \mathbf{1}\{\text{female}_i = g\} \cdot \pi_{kg} + \varepsilon_i,$$

so that $\hat{\pi}_{kg} = \widehat{\mathbb{E}}[\log w_i \mid \text{age}_i = k, \text{female}_i = g]$. Equivalently: age dummies, a female dummy, and *all* age $\times$ female interactions: 84 parameters for 84 cells. Again the model is saturated, so it is exactly non-parametric: the two curves in (e) are read straight off the coefficients.

(g) Now the parametric model

$$y_i = \beta_0 + \beta_1 \text{female}_i + \beta_2 \text{age}_i + \beta_3 \text{age}_i^2 + r_i.$$

Log daily wages, Veneto 2001, ages 19–60

	Log daily wage
female	−0.2663*** (0.0009)
age	0.0293*** (0.0004)
age ²	−0.0002*** (0.0000)
Constant	4.0578*** (0.0070)
Observations	699,902
R^2	0.171
Heteroskedasticity-robust (HC1) standard errors.	
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$	

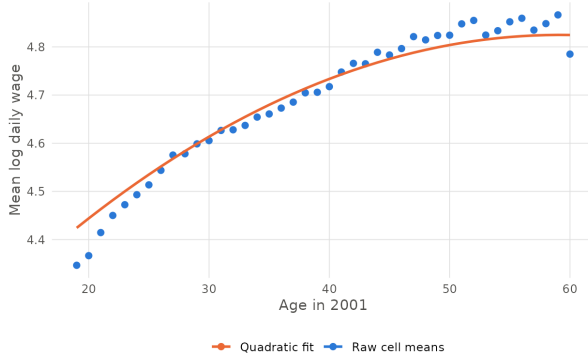
Interpretation of each coefficient:

- β_0 : the fitted mean log wage of a man at age = 0.
- $\hat{\beta}_1 = -0.2663$: the female differential, *restricted by the model to be constant at every age*. This is around 25% lower daily wages.
- β_2 and β_3 jointly govern the age profile, which is common to both genders:

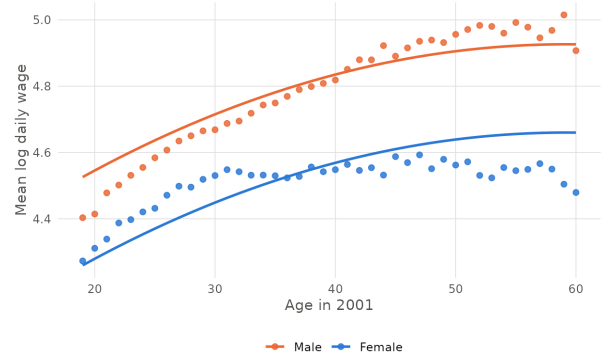
$$\frac{\partial \mathbb{E}[y_i \mid \text{female}_i, \text{age}_i]}{\partial \text{age}_i} = \beta_2 + 2\beta_3 \text{age}_i.$$

With $\hat{\beta}_2 = 0.0293 > 0$ and $\hat{\beta}_3 = -0.000248 < 0$ the profile is concave and peaks at $\text{age}^* = -\hat{\beta}_2/(2\hat{\beta}_3) = 59.2$. The implied return to a year of age is 0.0170 at age 25, 0.0095 at 40, and −0.0004 at 60.

(h) *Graphical test*. Overlay the saturated cell means $\hat{\pi}_{kg}$ from (f) on the profile implied by the quadratic model of (g).



Pooled



By gender

Raw cell means vs. quadratic model

The saturated model of (f) is exactly identified: one parameter π_{kg} per (age,gender) cell, one moment condition per cell. The parametric model instead maps the same moments into only four parameters:

$$\begin{aligned}\mathbb{E}[\log w_i \mid \text{female}_i = 0, \text{age}_i = k] &= \beta_0 + \beta_2 k + \beta_3 k^2, \\ \mathbb{E}[\log w_i \mid \text{female}_i = 1, \text{age}_i = k] &= \beta_0 + \beta_1 + \beta_2 k + \beta_3 k^2.\end{aligned}$$

With ages $19, \dots, 60$ and two genders there are $42 \times 2 = 84$ cells, hence

$84 - 4 = 80 \text{ over-identifying restrictions.}$

You can conduct a more formal test with a minimum distance estimator.

Question 5: Response Function and Propensity Score

(a) The alternative response function $Y(d) = \alpha + \beta D + U(d)$ is *less* restrictive because it lets the unobservable differ by treatment state. The implied treatment effect is

$$Y(1) - Y(0) = \beta + (U(1) - U(0)),$$

which is **heterogeneous across individuals**: β is now only the mean effect, since $\mathbb{E}[Y(1) - Y(0)] = \beta + \mathbb{E}[U(1) - U(0)]$. The original model imposes $Y(1) - Y(0) = \beta$ for *every* individual.

(b) Under the restrictive model $Y = \alpha + \beta D + U$, so

$$\mathbb{E}[Y \mid D, X] = \alpha + \beta D + \mathbb{E}[U \mid D, X] = \alpha + \beta D + \mathbb{E}[U \mid X],$$

Note also that

$$\mathbb{E}[D \mid X] = 1 \cdot \mathbb{P}(D = 1 \mid X) + 0 \cdot \mathbb{P}(D = 0 \mid X) = \mathbb{P}(D = 1 \mid X) = e(X).$$

Now apply the law of iterated expectations, conditioning on (D, X) :

$$\begin{aligned} \mathbb{E}[Y(D - e(X))] &= \mathbb{E}\left[\mathbb{E}[Y \mid D, X] (D - e(X))\right] \\ &= \mathbb{E}\left[(\alpha + \beta D + \mathbb{E}[U \mid X]) (D - e(X))\right] \\ &= \underbrace{\alpha \mathbb{E}[D - e(X)]}_{=0} + \beta \mathbb{E}[D(D - e(X))] + \underbrace{\mathbb{E}\left[\mathbb{E}[U \mid X] (D - e(X))\right]}_{=0} \\ &= \beta \mathbb{E}[D(D - e(X))]. \end{aligned}$$

■

(c) Under the model, realised outcomes satisfy

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0) = D_i (Y_i(1) - Y_i(0)) + Y_i(0) = \beta D_i + Y_i(0).$$

Substituting into the numerator of $\hat{\beta}$,

$$\frac{1}{N} \sum_i Y_i (D_i - \hat{e}(X_i)) = \beta \frac{1}{N} \sum_i D_i (D_i - \hat{e}(X_i)) + \frac{1}{N} \sum_i Y_i(0) (D_i - \hat{e}(X_i)),$$

so that

$$\hat{\beta} = \beta + \frac{\frac{1}{N} \sum_i Y_i(0) (D_i - \hat{e}(X_i))}{\frac{1}{N} \sum_i D_i (D_i - \hat{e}(X_i))}.$$

Assume (i) (Y_i, D_i, X_i) i.i.d. with finite second moments, (ii) $\hat{e}(X_i) \xrightarrow{p} e(X_i)$, and (iii) overlap, $0 < e(X) < 1$, so the denominator has a non-zero limit. Then by the LLN the ratio converges to $\mathbb{E}[Y(0)(D - e(X))] / \mathbb{E}[D(D - e(X))]$, and the numerator is zero:

$$\mathbb{E}[Y(0)(D - e(X))] = \mathbb{E}\left[\mathbb{E}[Y(0) \mid X] \underbrace{\mathbb{E}[D - e(X) \mid X]}_{=0}\right] = 0,$$

using $Y(0) \perp\!\!\!\perp D \mid X$. Hence $\hat{\beta} \xrightarrow{p} \beta$.

(d) Assume additionally $\mathbb{E}[U \mid X] = X' \gamma$. Write the (correctly specified) regression:

$$Y_i = \delta D_i + X_i' \gamma + U_i$$

By FWL (Q3)

$$\delta = E[\tilde{D}_i \tilde{D}_i']^{-1} E[\tilde{D}_i \tilde{Y}_i]$$

where the tilde objects are the residuals of the FWL regression on X_i . You can write

$$\begin{aligned} b &= \frac{E[\tilde{Y} \tilde{D}]}{E[\tilde{D}_i \tilde{D}_i']} = \\ &= \frac{E[(Y - E[Y|X])(D - E[D|X])]}{E[(D - E[D|X])^2]} \\ &= \frac{E[Y(D - E[D|X])] - E[E[Y|X](D - E[D|X])]}{E[D(D - E[D|X])] - E[E[D|X](D - E[D|X])]} \end{aligned}$$

By LIE:

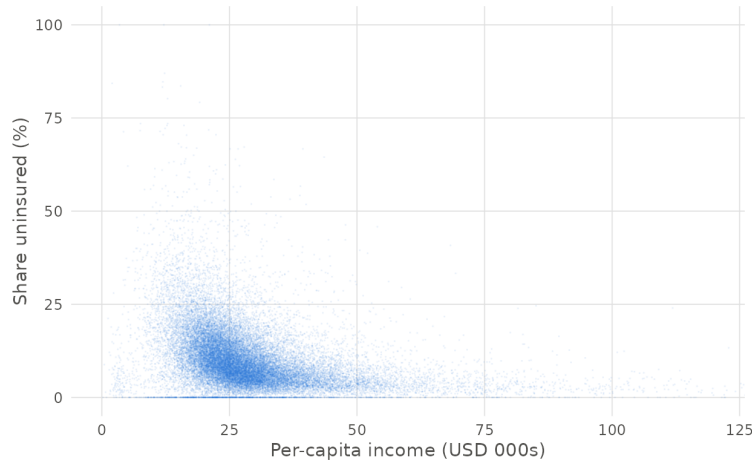
$$= \frac{E[Y(D - E[D|X])]}{E[D(D - E[D|X])]} = \beta$$

This model is more restrictive than the model $Y(d) + \alpha + \beta D + U(d)$ because besides the assumption $U \perp D|X$, we are imposing further assumptions to the structure of $E[U|X]$.

Question 6: On Binscatter

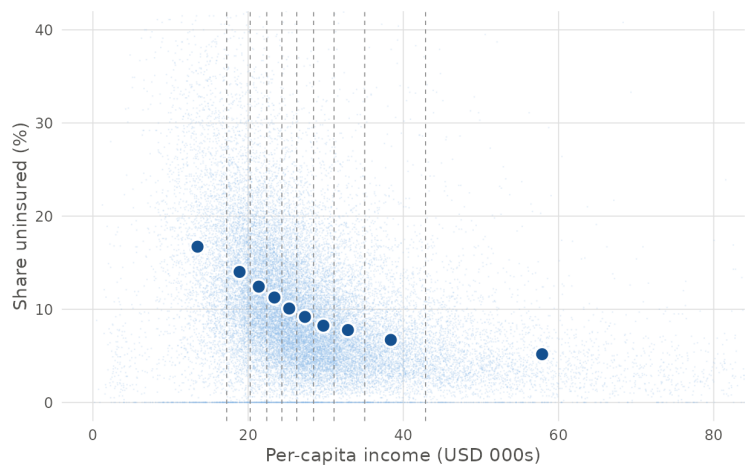
The question references the 2022 version of the paper. If you used the published 2024 version as reference, that is ok.

(a)



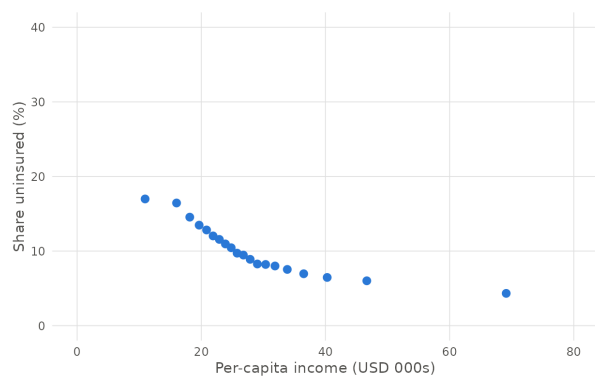
Share uninsured vs. per-capita income

(b) 1b in the 2024 version.

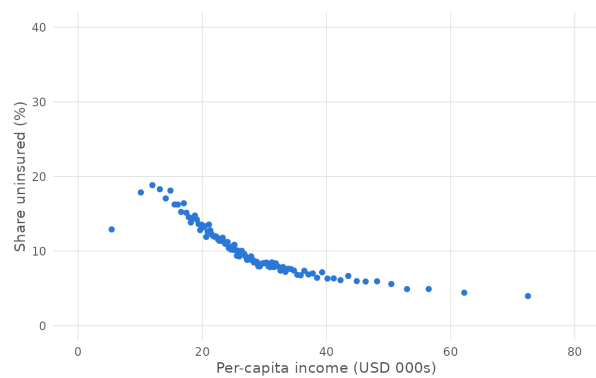


Deciles: canonical "manual" binscatter

(c)

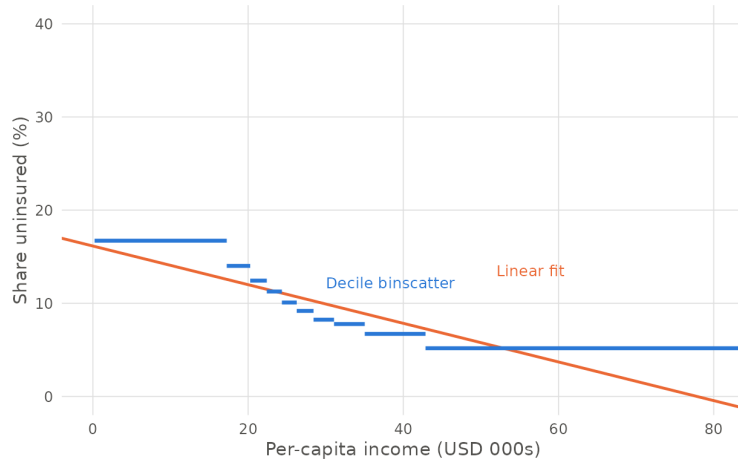


Vingtiles ($J = 20$)



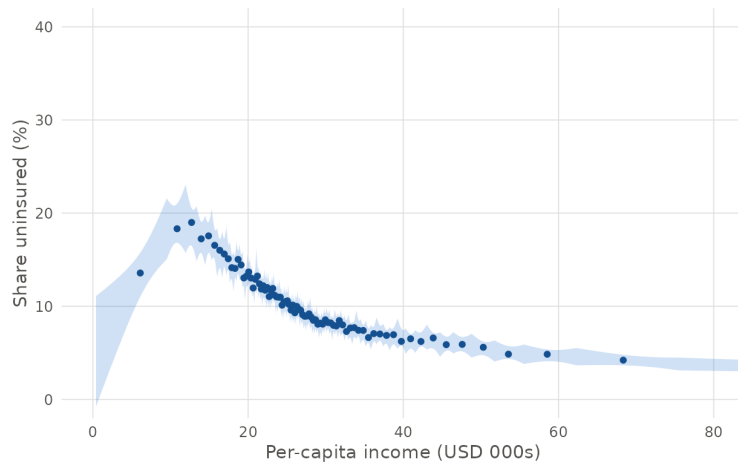
Centiles ($J = 100$)

(d) 1d in the 2024 version.



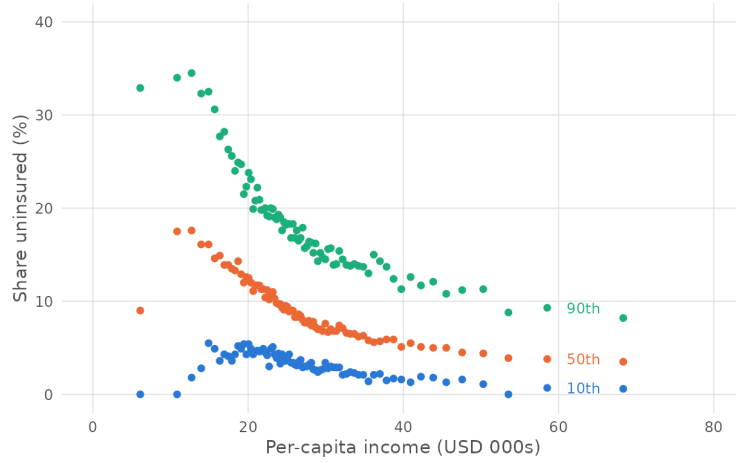
Deciles: piecewise-constant fit, with a linear fit through it

(e) 3a in the 2024 version



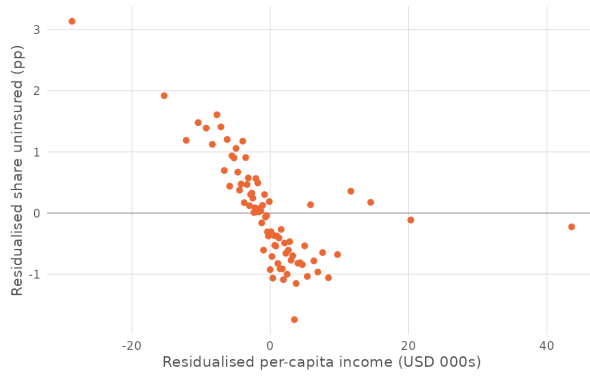
Cattaneo et al. optimal bin selector with uniform confidence bands

(f) Not in the 2024 paper. Conditional quantiles at 10th, 50th, 90th percentile. Same idea: in a given dollar value bin, extract the 10th, 50th, 90th percentile value and plot them against the bins.

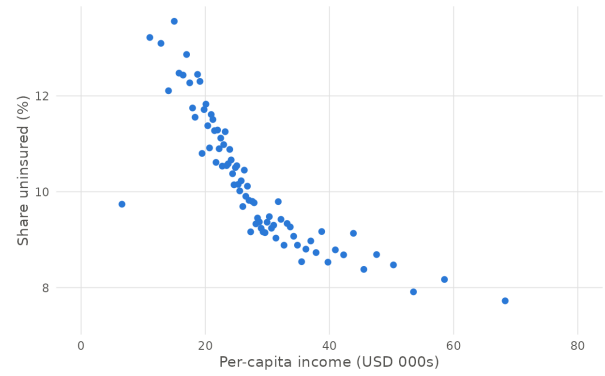


Quantile binscatter

(g) FWL: regress Y on W and X on W , keep the residuals $\tilde{Y}$ and $\tilde{X}$, and binscatter $\tilde{Y}$ against $\tilde{X}$.



(a) Residualisation / FWL



(b) Cattaneo et al.

Covariate-adjusted binscatter

Question 7: Contamination Bias in the STAR Experiment

(a) Keep students with non-missing `gkclasstype`: **6,325 kindergarteners in 79 schools** (small 1,900; regular 2,194; regular + aide 2,231). For each of the three kindergarten scale scores (`gkreadss`, `gkmathss`, `gkwordskillss`) compute percentile ranks **using only the 4,425 non-small-class students to define the reference distribution**.

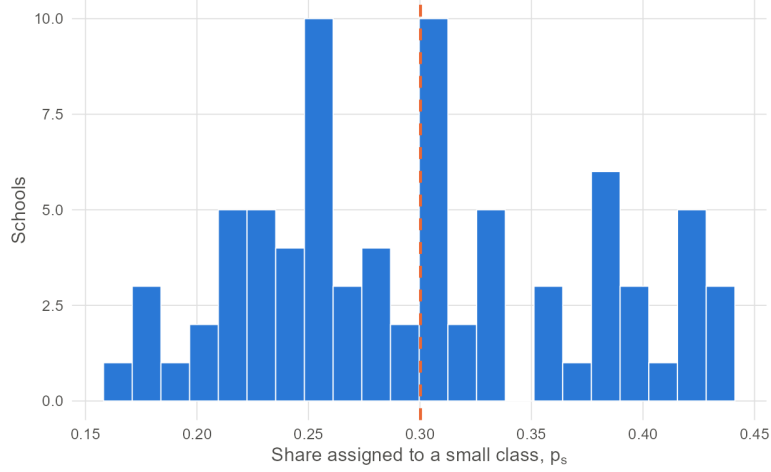
(b) Set

$$y_i = 100 \times \frac{1}{|M_i|} \sum_{m \in M_i} \text{rank}_{im}, \quad M_i = \{\text{read, math, word}\} \setminus \{\text{missing}\},$$

and y_i missing if $M_i = \emptyset$. This leaves **5,902 students in 79 schools**, with $\bar{y} = 51.4$ and $\text{sd}(y) =$

26.7.

(c) Randomisation is *within* school. Let $p_s = \mathbb{P}(D_i = 1 \mid \text{school}_i = s)$ be the share of school s 's kindergarteners assigned to a small class. It does vary: across the 79 schools $\hat{p}_s$ runs from 0.165 to 0.435.



Share assigned to a small class, by school

(d) We leverage the fact that conditional on school, assignment is random, so school fixed effects restore the identifying assumption $D_i \perp\!\!\!\perp (Y_i(0), Y_i(1)) \mid \text{school}_i$.

Does it identify a well-defined ATE? It identifies a well-defined *weighted* ATE, but not the ATE. School dummies are a *saturated* control, so the linear projection is the CEF: $\mathbb{E}^*[D \mid \text{school}] = \mathbb{E}[D \mid \text{school}] = p_s$. By FWL,

$$\beta^{OLS} = \frac{\mathbb{E}[\tilde{D}_i Y_i]}{\mathbb{E}[\tilde{D}_i^2]}, \quad \tilde{D}_i = D_i - p_{s(i)}.$$

Take the numerator and condition on the school first:

$$\mathbb{E}[(D_i - p_s)Y_i \mid s] = \text{Cov}(D_i, Y_i \mid s).$$

Now use $Y_i = Y_i(0) + D_i(Y_i(1) - Y_i(0))$ together with random assignment within school:

$$\begin{aligned} \text{Cov}(D_i, Y_i \mid s) &= \mathbb{E}[D_i Y_i \mid s] - p_s \mathbb{E}[Y_i \mid s] \\ &= p_s \mathbb{E}[Y_i(1) \mid s] - p_s \left(p_s \mathbb{E}[Y_i(1) \mid s] + (1 - p_s) \mathbb{E}[Y_i(0) \mid s] \right) \\ &= p_s(1 - p_s) \underbrace{\left(\mathbb{E}[Y_i(1) \mid s] - \mathbb{E}[Y_i(0) \mid s] \right)}_{\equiv \tau_s}, \end{aligned}$$

and the denominator is $\mathbb{E}[\text{Var}(D_i | s)] = \mathbb{E}[p_s(1 - p_s)]$. Therefore

$$\boxed{\beta^{OLS} = \sum_s \omega_s \tau_s, \quad \omega_s = \frac{\mathbb{P}(\text{school} = s) p_s(1 - p_s)}{\sum_{s'} \mathbb{P}(\text{school} = s') p_{s'}(1 - p_{s'})}} \quad (4)$$

(e) Estimating $y_i = \beta D_i + \sum_s \alpha_s \mathbf{1}\{\text{school}_i = s\} + \varepsilon_i$ with HC1 standard errors on all $N = 5,902$ students:

$$\hat{\beta} = 5.21 \quad (\text{s.e. } 0.69), \quad t = 7.6.$$

Assignment to a small class raises a student's average percentile rank by about **5.2 points of the non-small-class distribution**.

(f) Fitting the regression *separately* in each of the 79 schools: within a school the regression of y on a single binary D is saturated, so

$$\hat{\beta}_s \xrightarrow{p} \mathbb{E}[y_i | D_i = 1, s] - \mathbb{E}[y_i | D_i = 0, s] = \tau_s,$$

the school-specific effect of a small class relative to whatever the non-small students in that school were doing. Averaging with weights equal to school enrolment,

$$\hat{\beta}^{\text{avg}} = \sum_s \frac{N_s}{\sum_{s'} N_{s'}} \hat{\beta}_s \xrightarrow{p} \sum_s \frac{N_s}{\sum_{s'} N_{s'}} \tau_s = \mathbb{E}[\tau_i],$$

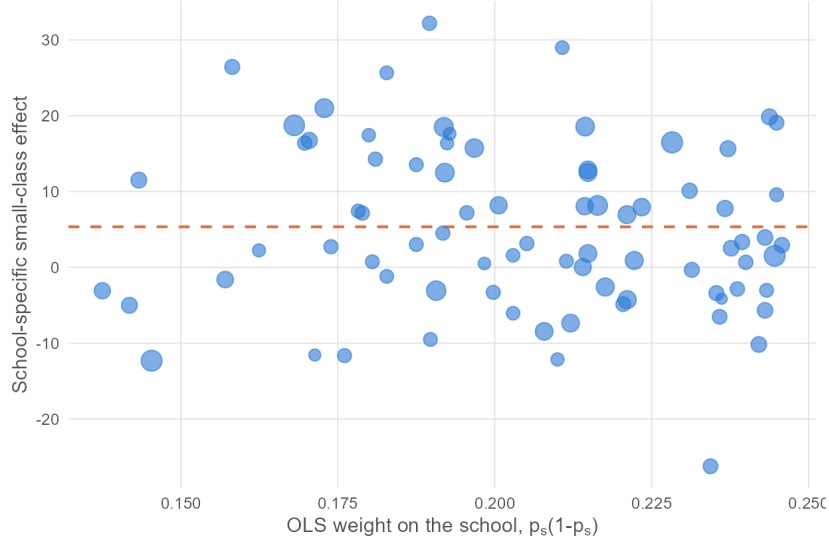
which *is* the ATE for the population of students in the experiment.

Comparing the two weighting schemes. Part (e) uses ω_s proportional to $N_s p_s(1 - p_s)$; part (f) uses ω_s proportional to N_s . I prefer (f): its estimand is the object we actually want to know. The price is precision: $\hat{\beta}_s$ can be noisier in schools with fewer students or p_s close to 0 or 1.

Results. The two numbers we have are

$$\hat{\beta}^{\text{avg}} = 5.35 \text{ (s.e. } 0.67) \quad \text{against} \quad \hat{\beta}^{OLS} = 5.21 \text{ (s.e. } 0.69).$$

The two estimands are close even though the school-specific effects are dispersed ($\text{sd}(\hat{\tau}_s) = 11.0$ percentile points, against an average effect of 5): the reason is that $\text{corr}(\hat{\tau}_s, p_s(1 - p_s)) = -0.13$, so the OLS weights are close to orthogonal to the treatment effects and re-weighting barely moves the average.



School-specific effects against their OLS weight

(g) Now the three treatment arms: small (D_{1i}) and aide (D_{2i}).

$$y_i = \tau_1 D_{1i} + \tau_2 D_{2i} + \sum_s \alpha_s \mathbf{1}\{\text{school}_i = s\} + \varepsilon_i. \quad (5)$$

The coefficient $\hat{\tau}_1$ no longer identifies a well-defined ATE. Goldsmith-Pinkham, Hull and Kolesár (2024) show that with multiple treatments the estimator decomposes as

$$\hat{\tau}_1 \xrightarrow{p} \underbrace{\sum_s \omega_s^{11} \tau_{1s}}_{\text{own effects, } \sum_s \omega_s^{11}=1} + \underbrace{\sum_s \omega_s^{12} \tau_{2s}}_{\text{contamination, } \sum_s \omega_s^{12}=0}.$$

The small-class coefficient (τ_1) loads on the *aide* effects. The source is that $\mathbb{E}^*[D_1 \mid \text{school}, D_2]$ is not the CEF (regression not saturated).

Estimates. Fitting (5) on all 5,902 students: $\hat{\tau}_1 = 5.31$ (s.e. 0.78) for the small class and $\hat{\tau}_2 = 0.21$ (s.e. 0.72) for the teaching aide.

(h) Regress D_{1i} on D_{2i} and school dummies. **This regression is not saturated.** The saturated version of $\mathbb{E}[D_1 \mid \text{school}, D_2]$ has one parameter per (school, D_2) cell: $79 \times 2 = 158$ parameters. The regression as written has $79 + 1 = 80$, imposing a common aide coefficient across schools.

The contamination weights are proportional to how far the linear projection is from the CEF. We can check how "bad" it is here by running the fully saturated regression and, for example, testing the joint null of interaction terms or by showing the improvement in the fit across the two regressions. In our case, the bias seems to be quite small, with modest improvements in the R squared from the inclusion of the school-specific slope.

(i) Repeat (f) with two arms: in each school fit

$$y_i = \tau_{1s}D_{1i} + \tau_{2s}D_{2i} + \alpha_s + \varepsilon_i,$$

then form $\hat{\tau}_1^{\text{avg}} = \sum_s (N_s / \sum N_{s'}) \hat{\tau}_{1s}$.

Yes, this identifies a well-defined ATE. Within a school the regression on two mutually exclusive arm dummies plus a constant is *fully saturated* in (D_1, D_2) (in schools where all arms are present). The linear projection equals the CEF and

$$\hat{\tau}_{1s} \xrightarrow{p} \mathbb{E}[y \mid D_1 = 1, s] - \mathbb{E}[y \mid D_1 = D_2 = 0, s] = \tau_{1s},$$

the school-specific ATE of small classes, with no aide contamination. These can be then aggregated like above. The same argument applies to τ_{2s} .

Results. 78 of the 79 schools have all three arms present (5,868 students):

$$\hat{\tau}_1^{\text{ATE}} = 5.56 \text{ (s.e. 0.76)}, \quad \hat{\tau}_2^{\text{ATE}} = 0.07 \text{ (s.e. 0.71)}.$$

(j) Of the three estimators Goldsmith-Pinkham, Hull and Kolesár propose, I use the **easiest-to-estimate weighted average (EW)** with their package *multe*.

Small classes and teaching aides in STAR kindergarten

	(1) PL <i>part (g)</i>	(2) ATE <i>part (i)</i>	(3) EW <i>part (j)</i>
Small class	5.31 (0.78)	5.56 (0.76)	5.30 (0.78)
Teaching aide	0.21 (0.72)	0.07 (0.71)	0.26 (0.71)
Students	5,902	5,868	5,868
Schools	79	78	78

Question 8: Attrition

(a)

- $E_{i1} = E_{i0} = 1$: the **always-employed**. They work whether or not they get the training.
- $E_{i1} > E_{i0}$: the **employment compliers**: not working absent training, working because of it.
- $E_{i1} < E_{i0}$: the **defiers**: training pushes them out of work.

- $E_{i1} = E_{i0} = 0$: the **never-employed**. No work, regardless of training status.

(b) *Constant effects*, $W_{i1} - W_{i0} = \beta$. This is a homogeneity assumption: the training raises *everyone's* log wage by the same amount.

Monotonicity, $E_{i1} \geq E_{i0}$. No defiers. Reasonable in a job training context.

(c) We run W_i on a constant and D_i among $\{E_i = 1\}$, so the coefficient is

$$b = \mathbb{E}[W_i \mid D_i = 1, E_i = 1] - \mathbb{E}[W_i \mid D_i = 0, E_i = 1].$$

By exogeneity, $\{D_i = 1\}$ is a random subset of the population, so conditioning on $D_i = 1$ and $E_i = 1$ is conditioning on $\{E_i(1) = 1\}$, which under monotonicity is the always-employed *plus* the compliers:

$$\mathbb{E}[W_i \mid D_i = 1, E_i = 1] = (1 - p_0) \mathbb{E}[W_{i1} \mid AT] + p_0 \mathbb{E}[W_{i1} \mid C],$$

where

$$p_0 \equiv \mathbb{P}(E_{i0} = 0 \mid E_{i1} = 1) = \frac{\mathbb{P}(E_{i1} > E_{i0})}{\mathbb{P}(E_{i1} = 1)}.$$

For the control arm, monotonicity says $\{E_i(0) = 1\}$ is the always-employed:

$$\mathbb{E}[W_i \mid D_i = 0, E_i = 1] = \mathbb{E}[W_{i0} \mid AT].$$

Now substitute $W_{i1} = W_{i0} + \beta$ for all groups:

$$\begin{aligned} b &= (1 - p_0)(\mathbb{E}[W_{i0} \mid AT] + \beta) + p_0(\mathbb{E}[W_{i0} \mid C] + \beta) - \mathbb{E}[W_{i0} \mid AT] \\ &= \beta + (1 - p_0)\mathbb{E}[W_{i0} \mid AT] + p_0\mathbb{E}[W_{i0} \mid C] - \mathbb{E}[W_{i0} \mid AT], \end{aligned}$$

so that

$$\boxed{b = \beta + p_0(\mathbb{E}[W_{i0} \mid C] - \mathbb{E}[W_{i0} \mid AT])} \tag{6}$$

The regression identifies β only if $p_0 = 0$ (the programme moved nobody into work) or the compliers and the always-employed have the same potential untreated wages.

(d) $p_0 = 0.10$ means: **one in ten of the workers we observe in the treatment arm would not have been working at all without the training.**

From (6) with $\beta = 0.05$,

$$b = 0.05 + 0.10 \times (\mathbb{E}[W_{i0} \mid C] - \mathbb{E}[W_{i0} \mid AT]),$$

so the answer depends on the sign of the gap in potential outcomes between the marginal workers and the inframarginal ones. That is an **economic** assumption, not a statistical one. The natural one

is a reservation-wage type argument: a person works when their wage offer clears their reservation wage. The always-employed clear it in both states; the compliers clear it only once training shifts their offer up. So the compliers are drawn from *just below* the threshold and the always-employed from above it, giving $\mathbb{E}[W_{i0} | C] < \mathbb{E}[W_{i0} | AT]$ and hence

$$b < 0.05.$$

(e) The extra assumption (the highest W_{i1} among compliers is below the lowest W_{i1} among the always-employed) is a *rank* restriction. It says the compliers are not just lower-paid on average but occupy the bottom of the treated wage distribution. Combined with $p_0 = 0.10$, we can place them: among $\{D_i = 1, E_i = 1\}$, the compliers are the bottom 10% of observed wages.

Let $q_{0.10}$ be the 10th percentile of $W_i | D_i = 1, E_i = 1$. Then

$$\mathbb{E}[W_{i1} | AT] = \mathbb{E}[W_i | D_i = 1, E_i = 1, W_i > q_{0.10}],$$

and from (c) the control arm already gives $\mathbb{E}[W_{i0} | AT] = \mathbb{E}[W_i | D_i = 0, E_i = 1]$. Hence

$$\boxed{\beta = \mathbb{E}[W_i | D_i = 1, E_i = 1, W_i > q_{0.10}] - \mathbb{E}[W_i | D_i = 0, E_i = 1]}. \quad (7)$$

(f) $Y_i = \exp(W_i) E_i$ is **earnings in levels, with a zero recorded for everyone not working**. It is a composite of the extensive and intensive margins and it is observed for *every* unit. The resulting ATE will be a dollar amount that combines changes in wages and employment. To evaluate its magnitude, it can be expressed as a percentage change of pre-treatment means or in relation to the cost of training.

(g) We can write the ATE:

$$E[Y_i | D_i = 1] - E[Y_i | D_i = 0] = E[E_{i1} e^{W_{i1}}] - E[E_{i0} e^{W_{i0}}]$$

And under monotonicity (no defiers and no contribution from never takers):

$$ATE = \mathbb{P}(AT)(E[1e^{W_{i1}} | AT] - E[1e^{W_{i0}} | AT]) + \mathbb{P}(C)(E[1e^{W_{i1}} | C])$$

Extensive margin. The employment effect is observed directly, $\pi \equiv \mathbb{P}(E = 1 | D = 1) - \mathbb{P}(E = 1 | D = 0) = \mathbb{P}(C)$, and its earnings contribution is $\pi \cdot \mathbb{E}[e^{W_{i1}} | C]$. From the assumption of part (e) we get:

$$\mathbb{E}[e^{W_{i1}} | C] = \mathbb{E}[e^{W_i} | D_i = 1, E_i = 1, W_i \leq q_{0.10}],$$

the mean earnings of the bottom $\hat{p}_0$ share of the treated workers.

Intensive margin. From $W_{i1} = W_{i0} + \beta$:

$$\mathbb{P}(AT) \left(\mathbb{E}[e^{W_{i1}} | AT] - \mathbb{E}[e^{W_{i0}} | AT] \right) = \mathbb{P}(E = 1 | D = 0) (e^\beta - 1) \mathbb{E}[e^{W_i} | D_i = 0, E_i = 1],$$

since $\mathbb{P}(AT) = \mathbb{P}(E_{i0} = 1) = \mathbb{P}(E = 1 | D = 0)$ and the wage piece is computable from in data assuming e). This allows to compute the decomposition: normalizing by the total effect gives shares interpretable as the relative "contribution" of each.